

Emergence of Specific *gyrA* Mutations Associated High-Level Fluoroquinolone-Resistant *Mycobacterium tuberculosis* among Multidrug-Resistant Tuberculosis Cases in North India

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Aim: This study aims to determine the frequency and pattern of *gyrA/B* mutations in multidrug-resistant (MDR) *Mycobacterium tuberculosis* (MTB) strains and also to assess the association between different *gyrA/B* mutations with phenotypic resistance to moxifloxacin (MOX) at clinical breakpoint (CB) drug concentration.

Method: A total of 106 clinical MTB isolates carrying *gyrA/B* mutations were included consecutively. Culture-based MOX susceptibility was tested at CB (1.0 µg/mL) followed by its correlation with *gyrA/B* mutations using Genotype MTBDRsl assay. The mutations associated with phenotypic resistance were further analyzed on a large dataset of 1,825 MDR tuberculosis (TB) patients.

Result: D94G and A90V mutations within *gyrA* were significantly associated with resistance and susceptible phenotype ($p < 0.001$), respectively. Of 1,825 MDR patients, *gyrA/B* mutations were found in 58.8% cases, of which fluoroquinolone (FQ) resistance was concluded among 97.9%, 0.8%, and 1.3% patients due to mutation in *gyrA*, *gyrB*, and in both the genes, respectively. D94G alone (45.9%) followed by A90V (21.2%) mutations in *gyrA* gene was most frequent.

Conclusion: Our study showed that MDR-TB has emerged in northern India with additional FQ resistance. Different selection pressure and transmission may result in prevailing accumulation of specific *gyrA* mutations causing high-level FQ resistance, therefore, current control measures need to be strengthened.

Keywords: *Mycobacterium tuberculosis*, fluoroquinolones, moxifloxacin, multidrug resistance

Introduction

GLOBAL EMERGENCE OF multidrug-resistant tuberculosis (MDR-TB; defined as resistance to a minimum of rifampin and isoniazid drugs) has threatened the efforts toward the goal of TB elimination. The treatment of MDR-TB is long and expensive, and it substantially lowers the success rates (57%) as compared with drug-sensitive TB (85%).¹ Fluoroquinolones (FQs), key antitubercular drugs, are recommended to be included in all regimens (unless contraindicated) for the treatment of drug-resistant tuberculosis.² FQ resistance among MDR-TB could be a serious issue because it may further worsen the treatment success rate. A timely and clinically reliable drug susceptibility testing (DST) is therefore crucial for effective treatment management. Culture-based phenotypic DST methods are currently the gold standard method that relies on the testing of *Mycobacterium tuberculosis* (MTB) strains at recommended

critical concentration (CC) of drug, which is specific for each anti-TB drug and test method.

Currently, the two later generation FQs, levofloxacin and moxifloxacin (MOX), are being used in the treatment regimen for MDR-TB. FQ resistance in MTB is mainly due to the acquisition of point mutations within the quinolone resistance-determining region of *gyrA* and *gyrB* genes, which encode DNA gyrase.³ A few specific codon mutations of *gyrA* (at position 94 other than D94A) were reported to be associated with high level of MOX resistance and may cause unfavorable treatment outcome in MDR-TB patients.^{4,5} Other mutations in *gyrA/B*, however, may cause only low-level FQ resistance and treatment with a higher dose of later-generation FQ-containing (up to 800 mg once daily) regimen may still be effective.⁶ Therefore, *in-vitro* culture-based susceptibility testing for MOX at the clinical breakpoint (CB; a concentration higher than CC and that can differentiate strains that will likely respond to treatment

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from those which will likely not respond to treatment) is recommended to design appropriate treatment regimen.^{7,8}

With increased understanding of the molecular basis of FQ resistance, new rapid and accurate diagnostic methods have developed. Genotype MTBDRs/ assay version 2.0 (Hain Lifescience, Nehren, Germany) is one among the few commercially available molecular detection assay, which is highly sensitive and specific to detect FQ resistance (by targeting *gyrA/B* genetic region) and resistance to second-line injectable drugs (SLIDs; by targeting *rrs* and promoter region of *eis* genes).⁹ In 2016, World Health Organization (WHO) recommended its use as an initial test, rather than phenotypic culture-based DST, to detect FQ and SLID resistance in confirmed MDR-TB patients.¹⁰ Genotypic assays targeting mutations related to drug resistance are important tools in molecular epidemiology. It helps to gain insight on disease transmission, effectiveness of drug-resistance management strategy, and it may also be used by TB epidemiology program to improve control and prevention strategies.

India has the highest burden of TB in the world. A high rate of FQ resistance has been reported among MDR-TB,¹¹ which is largely due to widespread and indiscriminate use of FQs in India.¹¹ Despite the fact that FQ resistance may contribute to unsuccessful treatment outcomes, comprehensive information on molecular epidemiology of FQ-resistant MTB remains lacking. In this study, we correlated the genotypic mutations in *gyrA/B* genes with phenotypic resistance at CB. Additionally, we analyzed the precise frequency and pattern of mutations associated with high-level FQ resistance on a large genotypic dataset of MDR-TB patients.

Materials and Methods

Study population

Under programmatic management of drug-resistant tuberculosis, our reference laboratory (accredited with ISO 15189) provides quality-assured services of genotypic and phenotypic DST for a wide geographical region of Uttar Pradesh, North India. For comparing phenotypic and genotypic resistance to FQs, we included 106 clinical MTB isolates that were recovered consecutively between February and March, 2019 from confirmed MDR or rifampicin-resistant (RR)-TB patients who had additional resistance to at least to FQs.

To know the frequency and diversity of different *gyrA/B* gene mutations, we retrospectively analyzed the test records of Genotype MTBDRs/ assay conducted in the year 2018. During this period, a total of 1,825 newly diagnosed MDR or RR-TB patients (before initiating the MDR-TB regimen) were tested at our laboratory and test results were included for analysis. Ethics approval of this study was not required for this study.

Genetic analysis of *gyrA* and *gyrB* genes

We used the commercial Genotype MTBDRs/ Assay Kit for detecting the mutation in *gyrA* and *gyrB* genes. This test was performed on processed sputum samples that involve DNA extraction, PCR-based amplification, and reverse hybridization of amplicon on DNA probe-labeled strip as per the manufacturer's instruction manual.¹² The strip contains 27 probes to check internal controls, identification of MTB

complex, and drug target regions *gyrA*, *gyrB* (for FQ resistance), *rrs* region (for SLIDs), and *eis* (for low-level kanamycin resistance) of MTB. The *gyrA* probes target codons 85–97 of the gene and *gyrB* probes target codons 536–541 of the gene.

After the test was completed, the DNA probe-labeled strip was cautiously interpreted. Briefly, the patterns comprising all wild-type (WT) bands without any mutant bands (MUT) were considered susceptible. Resistance was concluded if (i) one or more WT bands are absent, (ii) MUT bands, with or without the simultaneous absence of the complementary WT, are present.

Phenotypic liquid culture DST

Phenotypic DST was performed using the BACTEC MGIT 960 (BD, Sparks, MD). MTB inoculum for DST was recovered by inoculating the processed sputum in liquid culture media (BBL MGIT medium tubes; BD). Susceptibility testing of all 106 strains was tested for MOX at CB (1.0 µg/mL).

Data and statistical analysis

The proportions of susceptible and resistant strains harboring specific mutations were presented in percentage and compared using the chi-square test. A *p*-value of 0.05 was considered to indicate statistical significance.

Results

On analyzing the phenotypic DST of 106 *gyrA/B*-mutated MTB isolates, 61.3% of isolates showed MOX resistance at CB. Mutations at codon 94 and 90 positions, especially D94G and A90V, in *gyrA* gene were most frequent among these 106 isolates. However, their distribution between phenotypic susceptible and resistant strains varied. At CB, the distribution of D94G mutation alone was predominant ($p > 0.001$) in resistant isolates, whereas A90V mutation was distributed significantly ($p > 0.001$) among susceptible isolates (Table 1). In general, the distribution of all mutations at 94 position (or spanning region) of *gyrA* gene was higher among resistant strains. Other mutations, including those detected in *gyrB*, were displayed predominantly by susceptible isolates. The sensitivity of A90V (94.4%; confidence interval [CI] 0.04–8.28) followed by S91P (75.0%; CI 19.41–99.37) mutations in *gyrA* gene was highest to predict susceptible phenotype to MOX at revised CB. On another side, D94G mutation in *gyrA* gene had the highest sensitivity (70.77%; CI 58.70–81.40) to predict resistance phenotype at CB.

To assess the precise frequency of these clinically relevant *gyrA/B* mutations, we analyzed the test records of Genotype MTBDRs/ assay retrospectively. Of the total 1,825 MDR-TB patients tested, 1,073 (58.8%) were found to be FQ resistant with or without additional resistance to SLIDs. Of these, 1,051 (97.9%) had mutation in *gyrA* gene, 8 (0.8%) had mutation in *gyrB* gene, and 14 (1.3%) had mutation in both genes. The frequency of different mutations detected in *gyrA* and *gyrB* genes are shown in Table 2. About 91.0% of samples had known point mutations in the *gyrA*; the remaining 9.0% had unknown mutations in specific genetic regions of *gyrA* or *gyrB* genes. The point mutations, D94G (45.9%) and A90V (21.2%) in *gyrA* gene

TABLE 1. CORRELATION OF DIFFERENT MUTATIONS IN *gyrA* AND *gyrB* GENES WITH PHENOTYPIC SUSCEPTIBILITY PROFILE AT CLINICAL BREAKPOINT (1.0 µg/mL) DRUG CONCENTRATION

Gene	Mutation/wild-type probe	Probe band	Mutated codon/region	Mutation frequency (%)		p*
				MOX-susceptible (N=41)	MOX-resistant (N=65)	
<i>gyrA</i>	MUT probe	MUT 1	A90V	17 (41.46)	1 (1.54)	>0.001
		MUT 2	S91P	3 (7.32)	1 (1.54)	0.296
		MUT 3A	D94A	2 (4.88)	6 (9.23)	0.48
		MUT 3B	D94N/D94Y	0 (0)	1 (1.54)	1.000
		MUT 3C	D94G	15 (36.59)	46 (70.77)	>0.001
		MUT 3D	D94H	0 (0)	1 (1.54)	1.000
		MUT 1 and 3C	A90V; D94G	0 (0)	1 (1.54)	1.000
		MUT 2 and 3C	S91P; D94G	0 (0)	1 (1.54)	1.000
		MUT 3A and 3C	D94A; D94G	0 (0)	1 (1.54)	1.000
		MUT 3B and 3C	D94N/D94Y; D94G	0 (0)	2 (3.07)	0.521
	WT probe	WT 1	(85–89)	2 (4.88)	1 (1.54)	0.588
		WT 3	(92–96)	0 (0)	2 (3.07)	0.521
		WT 1 and 2	(85–89); (89–93)	0 (0)	1 (1.54)	1.000
		WT 2 and 3	(85–89); (92–96)	1 (2.44)	0 (0)	0.389
<i>gyrB</i>	WT probe	WT 1	(536–541)	1 (2.44)	0 (0)	0.389

*p-Value calculated comparing the distribution of individual mutation between phenotypic susceptible and resistant strains at CB drug concentration (1.0 µg/mL).

CB, clinical breakpoint; MOX, moxifloxacin; MUT, mutation; WT, wild-type.

were most frequent and caused FQ resistance (irrespective of the resistance level) in more than 2/3rd of our study population. There were only a minority of cases (4.6%) that had two or more mutations in *gyrA/B* genes.

Discussion

This is probably the first large-set data on molecular epidemiological of FQ-resistant MDR-TB from the country

that contributes highest number of drug-resistant TB in the world. Screening of newly diagnosed MDR-TB cases for second-line drugs using Genotype MTBDR_{sl} assay showed an alarmingly high rate (58.8%) of FQ resistance, which was in close agreement with our previous report.¹¹ Widespread and indiscriminate use of FQ and active transmission MTB in India is believed to be key contributor for such high resistance. Mutation in the *gyrA* gene was most common and it represents >99.0% patients who are considered as FQ

TABLE 2. FREQUENCY OF DIFFERENT GENETIC MUTATIONS IN *gyrA* AND *gyrB* GENES AMONG CONFIRMED MULTIDRUG-RESISTANT TUBERCULOSIS PATIENTS

Gene	MUT/WT probe n/N (%)	Probe band	Mutated codon/region	Pattern
				Frequency, n (%)
<i>gyrA</i>	MUT probe ^a 976/1,073 (91.0)	MUT 1	A90V	228 (21.2)
		MUT 2	S91P	57 (5.3)
		MUT 3A	D94A	70 (6.5)
		MUT 3B	D94N/D94Y	76 (7.1)
		MUT 3C	D94G	492 (45.9)
		MUT 3D	D94H	9 (0.8)
		MUT 1 and 3C	A90V and D94G	14 (1.3)
		MUT 3B and 3C	D94N/D94Y and D94G	8 (0.7)
		Others ^c	—	22 (2.1)
	WT probe ^b 75/1,073 (7.0)	WT 1	(85–89)	11 (1.0)
		WT 2	(89–93)	12 (1.1)
		WT 3	(92–96)	47 (4.4)
		Others ^d	—	5 (0.5)
<i>gyrB</i>	WT probe ^b 8/1,073 (0.7)		(536–541)	8 (0.7)
Mutation in both <i>gyrA</i> and <i>gyrB</i> ^e			14/1,073 (1.3)	14 (1.3)

Note: Isolates displaying mutation pattern with <0.5% of frequency were not listed in table.

^aPresence of mutation probe with or without corresponding wild-type probe.

^bAbsence of wild-type probe without appearance of corresponding mutation probe.

^cIncludes 15 different types of mutation pattern (in combination of ≥2 known mutations in *gyrA* gene).

^dIncludes three different types of mutation pattern (in combination of ≥2 unknown mutations in *gyrA* gene).

^eIncludes nine different types of mutation pattern (in combination of ≥2 mutations in *gyrA/B* genes).

resistant by molecular assay. In addition, most strains harboring *gyrB* mutations also had mutation in *gyrA* gene. Similar predominance of *gyrA* mutations has been documented in several previous studies from different countries.^{13–15} Likewise, in the panel of 106 *gyrA/B* mutated MTB isolates, D94G was most frequent and shown (alone or in combination with other mutations) by 49.3% of total 1,073 FQ-resistant patients (Table 2). High frequency of D94G mutations in our region posed serious concern as previous studies as well as our study on 106 *gyrA/B* MTB isolates showed high rates of FQ resistance in isolates harboring these mutations.^{7,16,17} Mutation at 94 codon position has also been associated with improved mycobacterial fitness. A study earlier showed the substantial proportion *Mycobacterium smegmatis* carrying double mutations in *rpoB* (associated with rifampicin) and *gyrA* genes had significantly higher fitness than the corresponding single mutants.¹⁸ In particular, the 94 codon mutation demonstrated greater fitness in all double mutants, regardless of the *rpoB* mutation. More than 62% of *gyrA/B* mutations in our study population were at 94 codon position, suggesting the evolutionary progression of MTB in our region, toward highly resistant genotypes.

Recent guidelines from the Global Laboratory Initiative-WHO indicate that all mutations in *gyrA* at 94 codon position (except D94A) are associated with high-level MOX resistance.¹⁹ Interestingly in our study, D94A mutation, which was earlier reported to cause a low-level MOX resistance,^{5,7} was displayed more by phenotypic resistant isolates (6 out of 8 isolates) than susceptible isolate at CB. Although D94A mutation was found in low frequency (6.5%; Table 2), phenotypic susceptibility at CB should be considered to detect false susceptibility so that the rate of treatment failure and disease transmission can be minimized.

Despite statistically significant distribution of D94G mutation among high-level MOX-resistant group of strains, a considerable proportion of strains (24.5%; Table 1) with D94G mutation alone showed MOX-susceptible phenotype (at 1.0 µg/mL). This indicates that D94G does not necessarily express into high-level MOX resistance and about 1/4th of MDR-TB isolates harboring D94G mutation may falsely be diagnosed as high-level MOX resistant. Since D94G mutation is most frequent among MDR-TB patients of our region (47.9%; Table 2), it seems that about 10% of MDR-TB patients with *gyrA/B* mutations are shifted to FQ-free treatment regimen, although they may get benefited with effective high-dose MOX-containing regimen.

Another most prevalent mutation in our study was A90V in *gyrA* gene and it was detected in 23.8% (256/1,073; Table 2) of FQ-resistant MDR-TB patients. Most of the isolates carrying A90V mutations alone (17 out of 18 isolates; Table 1) were phenotypically susceptible at CB, thus indicating good predictor ($p > 0.001$) for low-level MOX resistance. It is noteworthy that about 5% of strains with A90V mutations were found coexisting with a mutation at codon 94. The previous study stated that A90V mutation results in high-level resistance whenever it occurs concurrently with mutation at codon 94 (other than D94A).¹⁶

Previous functional studies showed that fluoroquinolones exert their action by binding to DNA gyrase enzyme, which is responsible for DNA supercoiling, thereby inhibiting DNA supercoiling of pathogens. Amino acid alterations at positions 90 and 94 in *gyrA* may lead to structural bumps or

disruption of water/metal ion bridge (critical for interaction of FQ with enzyme), thereby decreasing the drug affinity.²⁰

In a systematic analysis of preceding studies from different countries, both D94G and A90V mutations in *gyrA* were identified as predominant mutations among FQ-resistant isolates. On the contrary, most commonly reported mutation in a study from Iran, Portugal, Spain, and even from India were D94N (11%), S91P (42%), D84G (17%), D94A (20%), respectively.¹⁵ The discordant observation in these previous studies may be due to selection bias, different geographic location, or due to dynamic changes in FQ resistance. Consistent with our observation, a recent study from Northern India also identified D94G (61.9%) as most frequent followed by A90V (27.8%).⁹ Rest of other mutations in *gyrA* and *gyrB* genes were found in low frequency and their distribution between phenotypically susceptible and resistant (at CB) groups were statistically nonsignificant. Taking all the observation together, it appears that genotypic DST alone cannot exclude the possible high-level MOX resistance, thus additional testing by culture-based susceptibility test (at CB) would require to ensure better treatment outcomes.

Our study has some limitations. First, the susceptibility testing was performed at only one concentration of drug. Although, the selected drug concentration was clinically relevant, we could not correlate different mutations with a precise level of phenotypic resistance. Second, we could not exclude the possibility of FQ resistance influenced by resistance to other anti-TB drugs or mutations in drug determining genes other than *gyrA/B*. Besides mutations in *gyrA/B* gene, MTB may also adopt other strategy (like efflux pump) to confer FQ resistance. However, mutations in *gyrA/B* genes is a principal mechanism and it causes variable levels of resistance exclusively to fluoroquinolone class of drugs. Third, we only included 106 MTB isolates for comparing the genotypic and phenotypic FQ resistance profile. Although better evidence to predict the suitability of FQ-containing regimen would be welcome, our study clearly demonstrated the crisis related to FQ-resistance among MDR-TB patients and frequencies of clinically relevant *gyrA/B* mutations.

Conclusions

This study provides a precise frequency and pattern of *gyrA* and *gyrB* mutations (conferring variable level FQ-resistance) in MDR-MTB strains prevalent in a region of high-TB burden country. TB control program may use this epidemiological knowledge tactically to safeguard the efficacy of drug-resistant TB treatment regimens. MDR-TB has emerged as evolutionary fit by carrying *gyrA* mutations, pretreatment susceptibility testing, and cautious use of FQ-containing regimen is important for successful treatment management and to reduce the disease transmission. Furthermore, our study indicates that prediction of high-level FQ resistance by “Genotype MTBDRs/ assay” alone is questionable and therefore should be accompanied strategically with phenotypic testing at CB.

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